TinyML model details for ginger cultivation: four sub-modules with inputs, outputs, classes, alerts and preventive suggestions to framer.

Module 1 Soil moisture analysis & smart irrigation alerts:

Regression + rule-based threshold engine · Sensor input only · No camera

Soil & environmental input parameters:

Soil moisture (%) Capacitive sensor · 0–100%
Soil temperature (°C) DS18B20 · root-zone probe
Soil pH : Analog pH electrode
Electrical conductivity: EC sensor · mS/cm
Nitrogen (N) ppm/ Phosphorus (P) ppm / Potassium (K) ppm: NPK analog sensor
Ambient temperature: DHT22 · °C
Relative humidity: DHT22 · %
Rainfall (mm): Tipping bucket gauge
Days since irrigation: Firmware counter

Moisture level	Range (%)	Severity	Alert & preventive action
Critically dry	< 25%	Critical	Trigger irrigation relay immediately. Alert: "Soil critically dry — activate drip for 45 min." Suggest mulching to retain moisture. Check for wilting symptoms.
Moisture deficit	25–40%	Warning	Schedule irrigation within 6 hours. Alert: "Moisture low — irrigate before noon." Recommend 20–30 min drip cycle. Monitor next reading.
Optimal	40–70%	Normal	No irrigation needed. Alert: "Soil moisture optimal — no action." Continue monitoring every 2 hours.
Surplus moisture	70–85%	Warning	Suspend irrigation. Alert: "Excess moisture — risk of root rot." Open drainage channels. Reduce next irrigation cycle by 50%.
Waterlogged	> 85%	Critical	Immediate drainage required. Alert: "Waterlogging detected — risk of soft rot and bacterial wilt." Trigger drainage pump relay if present. Inspect rhizomes within 24 h.

Model output signals:
Moisture class label Irrigation ON/OFF relay signal Next irrigation time (minutes) pH alert flag NPK deficiency flag LED status colour

NPK & pH subsidiary alerts :

pH too low (< 5.5): "Acidic soil detected — apply lime at 200 kg/ha. Retest after 7 days."
pH too high (> 7.0): "Alkaline soil — apply sulphur or organic compost. Risk of micronutrient lockout."
Low N (< 80 ppm): "Nitrogen deficiency — apply urea 25 kg/ha or vermicompost. Yellowing expected in 5–7 days if untreated."
Low P (< 20 ppm): "Phosphorus low — apply SSP fertiliser. Affects root and rhizome development."
Low K (< 100 ppm): "Potassium deficiency — apply MOP. Risk of reduced disease resistance and poor rhizome fill."

Module 2 Pest infestation monitoring:

Image classification (MobileNetV2 INT8) + sensor fusion · Camera + sensors

Input — dual source:

Leaf image (camera): OV2640 · 96×96 px RGB
Stem image (camera): Close-up macro shot
Rhizome image: Ground-level camera view
Ambient humidity (%): Pest activity correlate
Temperature (°C): Pest lifecycle correlate
Days since last spray: Firmware log counter

Classification classes, symptoms & alerts:

Class	Visual symptoms	Severity	Alert & preventive suggestion
No pest	Clean leaf, no holes or discolouration	Normal	"No pest detected. Continue weekly monitoring." Schedule next inspection in 7 days.
Shoot borer (Conogethes punctiferalis)	Entry holes at pseudostem base, frass accumulation, wilting central shoot ("dead heart")	Critical	"Shoot borer detected. Remove and destroy infested shoots immediately. Apply Chlorpyriphos 0.05% or neem-based spray. Install pheromone traps 5 m apart."
Rhizome scale (Aspidiella hartii)	White/grey encrustations on rhizome surface, stunted shoot emergence	Critical	"Rhizome scale infestation. Treat seed rhizomes with Quinalphos 0.075% before planting. Remove heavily infested clumps. Avoid waterlogging."
Leaf roller (Udaspes folus)	Leaves rolled and bound with silk thread, feeding inside the roll	Moderate	"Leaf roller detected. Manually unroll and crush larvae. Spray Malathion 0.1% or release Trichogramma parasitoids as biological control."
Thrips (Scirtothrips dorsalis)	Silvery streaks on leaf surface, curling leaf margins, stippling	Moderate	"Thrips infestation. Apply blue sticky traps at canopy level. Spray Spinosad 0.01% or neem oil 3%. Increase humidity by light irrigation."
Mite infestation (Polyphagotarsonemus latus)	Bronze discolouration, fine webbing on leaf underside, leaf distortion	Moderate	"Mite damage detected. Spray Dicofol 0.05% or Spiromesifen. Avoid dusty field conditions. Increase irrigation frequency slightly to raise humidity."

Model output signals:

Pest class label
Confidence score (0–1)
Severity flag (Normal / Moderate / Critical)
Recommended pesticide
Spray schedule trigger
Alert to display / buzzer

Module 3 Disease detection — image-based TinyML

Image classification (EfficientNet-Lite INT8) · Camera input · 6 classes

Input:

Leaf image (camera): OV2640 · 96×96 px RGB
Pseudostem image: Base-level shot
Soil moisture (%): Disease risk correlate
Recent rainfall (mm): Fungal risk correlate
Humidity (%): Pathogen activity factor
Temperature (°C): Pathogen lifecycle factor

Disease classes, visual symptoms & outputs:

Class	Causative agent	Visual symptoms	Severity	Alert & preventive suggestion
Healthy	—	Deep green, erect leaves, no lesions or discolouration	Normal	"Plant healthy. Continue standard care. Next check in 7 days."
Bacterial wilt	Ralstonia solanacearum	Rapid yellowing & wilting of lower leaves, water-soaked pseudostem, foul smell when cut, milky ooze visible	Critical	"Bacterial wilt detected — most destructive ginger disease. Remove and destroy infected clumps immediately. Do NOT compost. Apply Copper oxychloride 0.3% as soil drench. Implement Anaerobic Soil Disinfestation (ASD) in affected plot. Quarantine the zone."
Soft rot	Pythium spp.	Water-soaked, mushy decay at stem base, foul odour, plant collapses suddenly	Critical	"Soft rot detected. Improve field drainage immediately. Remove infected plants. Apply Metalaxyl + Mancozeb 0.2% as soil drench. Avoid overwatering. Treat seed rhizomes with Trichoderma harzianum before next planting."
Leaf spot	Phyllosticta zingiberi	Circular to oval brown spots with pale centres and yellow halos, mainly on older leaves	Moderate	"Leaf spot detected. Remove severely affected leaves. Spray Mancozeb 0.25% or Copper oxychloride 0.3% at 10-day intervals. Improve air circulation by thinning crowded plants."
Rhizome rot	Fusarium oxysporum	Yellowing and browning at pseudostem base, internal rhizome tissue shows brown discolouration, hollow when squeezed	Critical	"Rhizome rot detected. Uproot and inspect rhizomes. Destroy infected material. Drench soil with Carbendazim 0.1% + Trichoderma. Do not re-plant ginger in same plot for 2 seasons."

Class	Causative agent	Visual symptoms	Severity	Alert & preventive suggestion
Chlorosis / nutrient deficiency	N, Fe, or Mg deficiency	Uniform yellowing (interveinal or whole leaf), no lesions, no water-soaked tissue	Moderate	"Chlorosis detected — likely nutrient deficiency, not disease. Cross-check NPK sensor. Apply ferrous sulphate 0.5% foliar spray for iron deficiency. Apply magnesium sulphate 1% for Mg deficiency. Recheck soil pH — alkaline pH locks out micronutrients."

Model output signals:

Disease class label
Confidence score (0–1)
Severity flag
Fungicide / bactericide recommendation
Quarantine zone flag
OLED display message
Buzzer / LED alert colour

Module 4 Yield estimation model:

Regression (Random Forest → C++ via micromlgen / TFLite INT8) · Sensor + derived inputs

Input parameters (collected across growing season):

Avg soil moisture (%): Rolling 30-day average
Avg soil temperature: Rolling 30-day average
Soil pH (current): At harvest stage
N, P, K levels: ppm at planting & mid-season
Cumulative rainfall: mm · season total
Disease alert count: Integer · from Module 3
Pest alert count : Integer · from Module 2
Irrigation events: Count · firmware log
Days since planting : Growth stage proxy
Planting density: Rhizomes/m² · manual input
Variety / cultivar: Categorical · encoded
Mulching applied: Binary flag · yes/no

Yield output classes & farmer actions:

Yield forecast band	**Estimated output**	**Interpretation**	**Recommended action**
Low yield	< 100 q/ha	Below avg	"Low yield predicted. Review disease/pest log. Apply top-dress fertiliser (K-rich). Consider extending growing period by 2–4 weeks if plants still green. Plan reduced harvest area."
Average yield	100–150 q/ha	Average	"Average yield expected (~125 q/ha). Standard harvest at 8–9 months. Arrange storage and market logistics for current stock levels."
Good yield	150–200 q/ha	Above avg	"Good yield predicted. Plan harvest in week 8.5–9. Arrange additional labour and storage capacity. Consider value-added processing (dry ginger / oil extraction)."
High yield	> 200 q/ha	Excellent	"Excellent yield predicted. Alert market aggregators in advance. Prioritise cold storage. Consider phased harvest to avoid market glut and price crash."

Model output signals:

- Predicted yield (q/ha) — point estimate
- Yield band (Low / Avg / Good / High)
- Confidence interval (±)
- Top yield-limiting factor
- Recommended harvest window (days)
- Market readiness flag

Regression approach detail:

Baseline: Always report the naive mean predictor (historical district average ~125 q/ha) as a baseline. Your model's value is measured against this.

Primary model: Random Forest Regressor (50 trees, max depth 8) — trained offline in Python (scikit-learn), exported to C++ arrays via micromlgen for ESP32 on-device inference. Suitable for small datasets (40–200 farm records).

Secondary model: TensorFlow Lite feedforward network (2 hidden layers, 32 neurons each, ReLU, INT8 quantized) — better when dataset grows beyond 500 records and non-linear interactions increase.

Evaluation metrics to report: RMSE (q/ha), MAE (q/ha), R^2 score, and comparison against baseline RMSE.